

Chapter 1: The Nature of Probability and Statistics

Statistics:

Statistics is the science of conducting studies to collect, organize, summarize, analyze, and draw conclusions from data.

Variable: A variable is a characteristic or attribute that can assume different values. Data are the values (measurements or observations) that the variables can assume.

Population and sample: A **population** is the entire group that you want to draw conclusions about. A **sample** is the specific group that you will collect data from. The size of the sample is always less than the total size of the population.



Statistics is sometimes **divided into two main areas**, depending on how data are used. The two areas are:

1. Descriptive statistics
2. Inferential statistics

1. Descriptive statistics: Descriptive statistics consists of the collection, organization, summarization, and presentation of data. In descriptive statistics the statistician tries to describe a situation.

Example: Consider the national census conducted by the U.S. government every 10 years. Results of this census give you the average age, income, and other characteristics of the U.S. population. To obtain this information, the Census Bureau must have some means to collect relevant data.

Once data are collected, the bureau must organize and summarize them. Finally, the bureau needs a means of presenting the data in some meaningful form, such as charts, graphs, or tables.

2. Inferential statistics: Inferential statistics consists of generalizing from samples to populations, performing estimations and hypothesis tests, determining relationships among variables, and making predictions.

EXAMPLE 1-1 Descriptive or Inferential Statistics

Determine whether descriptive or inferential statistics were used.

- a. The average jackpot for the top five lottery winners was \$367.6 million.
- b. A study done by the American Academy of Neurology suggests that older people who had a high caloric diet more than doubled their risk of memory loss.
- c. Based on a survey of 9317 consumers done by the National Retail Federation, the average amount that consumers spent on Valentine's Day in 2011 was \$116.
- d. Scientists at the University of Oxford in England found that a good laugh significantly raises a person's pain level tolerance.

SOLUTION

- a. Descriptive statistics were used because this is an average, and it is based on data obtained from the top five lottery winners at this time.
- b. Inferential statistics were used since this is a generalization made from a sample to a population.
- c. Descriptive statistics were used since this is an average based on a sample of 9317 respondents.
- d. Inferential statistics were used since an inference is made from a sample to a population.